

Computability: Exercise 4

Due: May 4, 2026

Question 1

Let $\Sigma = \{a, b\}$. show a TM that, given a word $w = w_1w_2 \cdots w_n$, outputs the word $w^R \stackrel{\text{def}}{=} w_nw_{n-1} \cdots w_1$. The output should be written on the tape when the machine halts, with nothing else written on the tape but blanks. Specify what working alphabet you use, and draw a full diagram of its states and transitions. Explain briefly how your construction works and the role of each of the states, but there is no need to prove correctness formally.

Question 2

Describe a TM that receives two binary numbers separated by $\#$, and calculates their sum (stops when their sum is written on the tape). For example, for the input $10\#11$ the output should be 101 .

Here there is no need to specify states and transitions, instead you may use a pseudo code that consists of simple procedures (such as “go to the beginning of the input”) that a Turing Machine can obviously perform based on our experience so far.

However, the running time of the TM (the number of computation steps until its run finishes) should be *polynomial in the number of letters in the input*. You may define the tape alphabet Γ as you wish.

briefly explain why your algorithm performs the required task, and why its running time is polynomial in the input size.

You may assume that the input is of the form described in the question (no need to validate it).

Question 3

Prove that any DFA A can be transformed into a TM M such that $L(A) = L(M)$ and M halts on every input.

In this question you should define *explicitly* what are $\Sigma, \Gamma, Q, q_0, F, \delta$.

Question 4

Describe a TM that gets as input a word of the form a^n , and outputs the binary representation of n . Try to make your machine as efficient as possible and analyze its running time. For full score, the running time should be $O(n \log n)$.

Question 5

Describe a TM that decides the language L over $\Sigma = \{a, b\}$, defined by

$$L = \{w \in \Sigma^* : \#_a(w) = \#_b(w)\}$$

Try to make your machine as efficient as possible and analyze its running time.